

Classwork 09//Markov Decision Process, Remedial for Quiz 3

Student Name: _____ Points: 30

A Number: _____ Due Date: April 14, 2026

1 Markov Decision Process (10 Points)

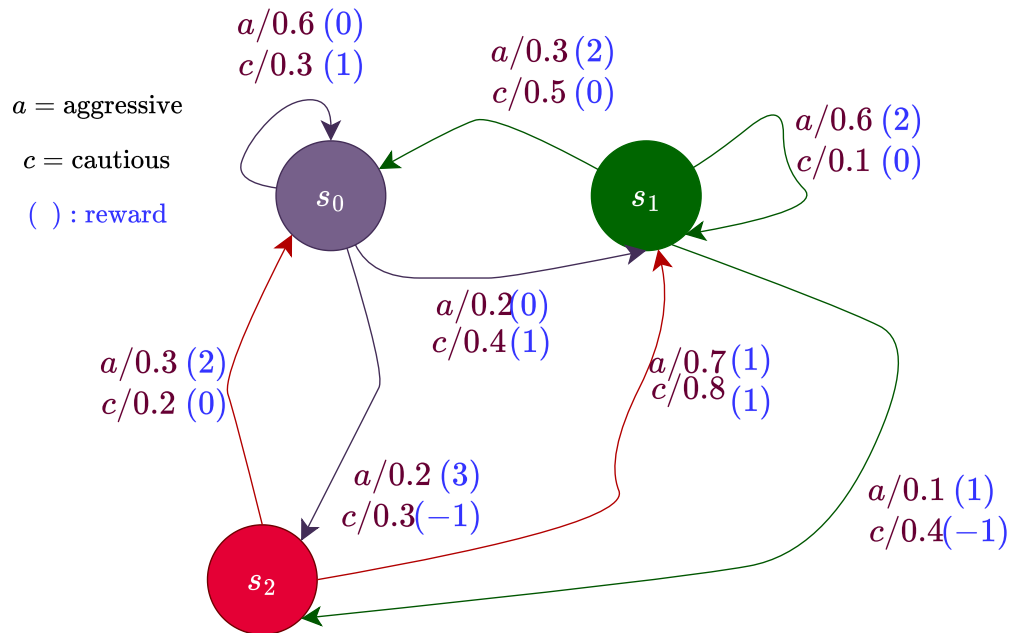


Figure 1: Q1

1. Expected immediate reward is given as

$$\bar{R}(s, a) = \sum_{s'} P(s'|s, a) R(s, a, s')$$

Calculate $\bar{R}(s_0, a)$ and $\bar{R}(s_0, c)$. Show your work.

(5 Points)

2. Based solely on expected immediate reward, which action (aggressive or cautious) is preferred in each state? Does the action stay consistent across all three states? **(5 Points)**
3. Suppose the agent follows the always aggressive policy π_a starting from s_1 . What is the expected reward after exactly one step? Repeat the same for always-cautious policy π_c . **(5 Points)**
4. Complete the transition table based on the diagram: **(5 Points)**

From	Action	To	Probability	Reward
s_0	a	s_0	0.6	0
s_0	a	s_1	0.2	0
s_0	a		0.2	3
s_0	c	s_0	0.3	
s_0	c		0.4	1
s_0		s_2	0.3	-1
s_1	a	s_0	0.3	2
s_1	a		0.6	2
s_1	a	s_2	0.1	1
s_1		s_0	0.5	0
s_1	c	s_1	0.1	0
s_1		s_2	0.4	-1
s_2	a	s_0	0.3	
s_2	a		0.7	1
s_2	c	s_0	0.2	0
s_2	c	s_1		1

Table 1: Full transition table. a = aggressive, c = cautious. Each row encodes $P(s' | s, a)$ and $R(s, a, s')$.

2 Quiz 3 Remedial

Please go through the solution of Quiz 3 posted on Canvas, and rework on the solution to every problem, and provide justification for the each of the solution. **(10 Points)**